
Abstract

Background: Cervical cancer, primarily linked to sexually transmitted human papillomavirus (HPV) infection, is a global health concern with approximately 604,000 new cases in 2020, ranking it as the fourth most common cancer worldwide. Prior clinical trials have shown high efficacy of the HPV vaccine in reducing cervical cancer precursors, particularly among younger populations, but its effectiveness in older populations remains less understood.

Objective: This study aims to investigate the association between HPV vaccination and cervical cancer in U.S. women aged 20–59 years.

Methods: Data was obtained from the National Health and Nutrition Examination Survey (2011-2020). Frequency distribution and descriptive statistics analysis were used to explore the participants' demographics. Bivariable and multivariable logistic regression was used to calculate the unadjusted and adjusted odds ratios in the association between HPV vaccination and cervical cancer, adjusting for age, race, education level, annual family income, and ever-been pregnant variables.

Results: Among the 8596 participants included in our sample, 98 (1.14%) participants self-reported being diagnosed with cervical cancer, in which 6 out of 1180 (0.5%) were vaccinated, and 92 out of 7416 (1.2%) were unvaccinated. After adjustments for sociodemographic characters, women who were vaccinated for HPV were less likely to be diagnosed with cervical cancer with OR = 0.4 (95% CI: 0.06, 1.47). However, these results are non-statistically significant.

Interpretation: Being vaccinated for HPV is significantly associated with lower risks of cervical cancer. Given this association, HPV vaccination should be encouraged for both children and adults to prevent HPV-related cancer cases.

Introduction

Cervical cancer, predominantly caused by sexually transmitted human papillomavirus (HPV) infection, represents a significant burden of disease worldwide⁽¹⁾. It is the fourth most common cancer, with 604,000 new cases in 2020 alone⁽²⁾. While risk factors such as age, education, ethnicity, smoking, socioeconomic status, multiple sexual partners, low immune status (e.g., HIV infection), and contraceptive pills contribute to the risk of cervical cancer, HPV plays a significant role in its development⁽³⁾. Notably, seven high-risk HPV types (16, 18, 31, 33, 45, 52 and 58) are causally associated with 90% of cervical cancer cases⁽⁴⁾.

The US approved first-generation quadrivalent HPV vaccines in 2006 targeting high-risk HPV types⁽⁵⁾. The second-generation nonvalent HPV vaccine was approved in 2016, which

additionally protects against HPVs not covered by the first-generation vaccines⁽⁶⁾. The HPV vaccine reduces the risk of invasive cervical cancer by 88% among women who were inoculated with the quadrivalent vaccine before the age of 17⁽⁷⁾, and the nonvalent vaccine has further proven to prevent an additional 18.3-20% of cervical cancers⁽⁶⁾. This effectiveness is attributed to the lower likelihood of HPV exposure before first sexual contact. Therefore, vaccine administration for age groups 11-16 years is the most efficient. However, vaccination for adults aged 27–45 shows some benefit in delaying the onset of certain cancers⁽⁸⁾.

The development of cervical cancer from HPV infection typically occurs over a span of 15 to 20 years (typically around the age of 50 years). However, several factors (e.g., immune status, the type of HPV infection, use of hormonal contraceptives, presence of other STIs, or the number of childbirths) can accelerate this process to 5-10 years^(2, 9). Given that HPV vaccinations were first administered in 2006, cervical cases are expected to emerge within this study's timeline between 2011 and 2020 if individuals were infected with HPV after receiving the vaccine.

Prior randomized clinical trials have demonstrated the vaccines' high efficacy in reducing cervical cancer precursors by 93% [95% CI: 87, 96]⁽¹⁰⁾, providing promising avenues for primary prevention. However, previous studies focus on the effectiveness of the vaccine on younger populations (e.g., adolescents and young adults), whereas its impact on older women is not as well understood. While post-licensure studies have primarily focused on HPV infection rates, limited data exist on the population effects of HPV vaccination, specifically against cervical cancer. Amid prevalent vaccine hesitancy⁽¹¹⁾ and a need for comprehensive evidence, this study aims to bridge the gap by investigating the association between HPV vaccination and cervical cancer in US women aged 20–59, leveraging publicly available NHANES datasets.

Methods

Study Design and Data Sources

The National Health and Nutrition Examination Survey (NHANES) is a continuous survey that involves interviews and physical examinations to evaluate the health and nutritional status of non-institutionalized US adults and children. NHANES is managed by the National Center for Health Statistics (NCHS), a part of the Centers for Disease Control and Prevention (CDC). In this cross-sectional study, we analyzed four data cycles (2011-2020). NHANES uses a nationally representative sample collected bi-annually using a complex multistage probability sampling design to achieve data representative of civilian non-institutionalized US adults. A detailed description of the sampling design and methods is available from NHANES⁽¹²⁾ and elsewhere⁽¹³⁾.

Analytic Sample

The sampling frame for the current analysis between HPV vaccination status and cervical cancer prevalence included US women aged 20-59 years with valid responses (Yes/No) to HPV vaccination status and cervical cancer. Participants younger than 20 or older than 59 years were

excluded because they were not eligible to answer questions about cancer status and HPV vaccination history. Responses “Refused,” “Don’t know,” or “Not stated” to the explanatory and outcome variable were also treated as missing and excluded from the analysis.

Overall, 45,464 participants completed the health and dietary questionnaire of NHANES during the four cycles. The final complete study sample included 8596 participants from 2011 to 2020. For more details about the selection process of the study sample, see Figure 1.

Study Variables

The dichotomous outcome variable for cervical cancer (yes/no) was obtained through two survey questions: “Ever told you had cancer or malignancy?” and “What kind of cancer?”. Previous research on self-reported cancer diagnosis indicates high sensitivity for the presence of disease, up to 80%⁽¹⁴⁾.

The explanatory variable of HPV vaccination status was obtained through the survey question “received HPV vaccine”; respondents were categorized as vaccinated or unvaccinated. A previous study has shown that the sensitivity of self-reported HPV vaccination status is high among adult women aged 18-49 years in the US, with 91%⁽¹⁵⁾.

Potential confounders include age, race (Asian, Black, Mexican American, White, and Other, including other Hispanic and multiracial), education level (12th grade or less, high school graduate/GED or equivalent, some college or AA degree, college graduate or above), annual family income (under \$20,000, \$20,000 to \$54,999, \$55,000 to \$99,999, \$100,000 and Over), and ever pregnant (yes/no).

Analysis Plan

Ethics approval for this study was covered under item 7.10.3 of the University of British Columbia’s policy number 89: Research involving human participants, a clause governing publicly available datasets⁽¹⁶⁾. All data management and analysis were conducted using R version 4.3.2. First, a frequency distribution descriptive statistics analysis was conducted to explore the participants’ demographics. Second, a multivariable logistic regression model was fitted to investigate the relationship between cervical cancer and HPV vaccination status, adjusting for age, race, education level, annual family income, and ever been pregnant.

Results

The final analytic sample consisted of 8,596 female participants with an average age of 39.54 years. The racial distribution revealed “White” as the predominant group (32.65%), while “Mexican American” was the least common (13.67%). Education levels varied, with the majority having some college or an associate’s degree (35.32%). Income ranged from under \$20,000 to

over \$100,000, with the majority of participants falling in the \$20,000 to \$54,999 range (22.55%). Notably, 66.49% of participants reported a history of previous pregnancies (Table 1).

As seen in Table 1, the vaccinated group had younger individuals with a median of 25 years, while unvaccinated were older with a median of 42 years ($p < 0.001$). The distribution of race ($p < 0.001$) and income ($p < 0.001$) among both groups were similar. In terms of education ($p < 0.001$), vaccinated individuals tend to have higher levels of education, with the majority having some college or associate's degree or above (73.5% in the vaccinated vs 61.9% in the unvaccinated group). Lastly, there was a higher percentage of women who were ever pregnant in the unvaccinated than in the vaccinated group (70% vs 48%, respectively, $p < 0.001$).

Table 2 shows the counts and frequencies of cervical cancer among the unvaccinated and vaccinated groups. The vaccination prevalence in the study sample was remarkably different, with 6.2% among those diagnosed with cervical cancer and 13.8% among those who were not diagnosed with cervical cancer. Overall, 1.14% (98/8596) of participants reported being diagnosed with cervical cancer; however, only 6 (0.5%) individuals reported being vaccinated, while 92 (1.2%) reported not having been vaccinated for HPV (Table 2).

As summarized in Table 3, the unadjusted bivariable logistic regression analysis (Model 1) gave an odds ratio (OR) of 0.41 [95% CI: 0.16, 0.85] ($p = 0.03$), suggesting a significant association between HPV vaccination and cervical cancer. Upon adjusting for potential confounders (Model 2), including age, education, race, income, and ever been pregnant, the model gives an OR of 0.4 (95% CI: 0.06, 1.47)); however, $p\text{-value} > 0.05$, indicating a non-statistically significant effect of HPV vaccination on the risk of developing cervical cancer (Table 3).

Discussion

Key Findings

This study's results suggested that HPV vaccination is associated with reduced risks of cervical cancer, which aligns with prior studies which found the vaccine's effectiveness in lowering the risk of cervical cancer among younger individuals⁽⁷⁾ and the potential to reduce risk in adults up to 45 years old⁽⁸⁾. For instance, a population-based study from Sweden⁽⁷⁾ utilized comprehensive registry data to compare HPV-associated cervical cancer incidence rates between vaccinated and unvaccinated populations and found a reduction in incidence rate ratios among vaccinated individuals [IRR = 0.51 (95% CI 0.32; 0.82)], even after adjusting for covariates [IRR = 0.37 (95% CI, 0.21; 0.57)], indicating a strong protective effect of HPV vaccination.

After adjusting for age, education, income, and pregnancy history, our analysis did not yield sufficient evidence to reject the null hypothesis of no association between HPV vaccination and cervical cancer. This indicates that age, education, income, and ever been pregnant do not act as

confounding factors in the relationship between HPV vaccination and cervical cancer, which is inconsistent with prior research. Previous studies suggest that increased age⁽¹⁸⁾, lower education level⁽¹⁹⁾ and socioeconomic status⁽²⁰⁾, and previously having full-term pregnancies⁽²¹⁾ increase the odds of developing cervical cancer, as well as decrease the likelihood of receiving HPV vaccines. For example, a prior study in Ecuador found women with primary education or less to have almost twice the cervical cancer incidence than those with secondary or higher education, whereas those who were illiterate had almost six times the incidence found among university-educated women⁽¹⁹⁾. Furthermore, a Korean study found that cervical cancer is more likely to be diagnosed in populations with lower income levels (OR, 1.46; 95% confidence interval [CI], 1.28 to 1.67 for the lowest compared to the richest)⁽²⁰⁾.

The non-significant association between HPV vaccination and cervical cancer can be explained by herd immunity. A previous study had shown a massive decline in HPV infections, from 22.7% in 2005–2007 and 7.3% in 2010–2012 to 1.5% in 2015; aPR = 0.40 (95% CI, 0.25–0.63) and 0.08 (95% CI, 0.02–0.26), respectively, compared with the prevaccine sample among unvaccinated people due to herd immunity from those vaccinated⁽²²⁾. In other words, due to the implementation of HPV vaccines, individuals who have not received vaccination also experience a reduced risk of contracting HPV and, thus, cervical cancer.

Further large-scale longitudinal research is needed, including more factors such as lifestyle factors and HIV status, by linking more data sources such as cancer registries and vaccination records. Although the efficacy and efficiency of the HPV vaccine on cervical cancer are well established^(10,23,24), this study can help extend knowledge about the HPV vaccine's effectiveness against cervical cancer in older US adults, evidence of which is currently lacking. Given HPV vaccination rates are relatively low in the US⁽²⁵⁾, findings from this study can be used by policymakers and health professionals to encourage further HPV vaccination uptake among adolescents aged 11–15 years and to support decision-making about HPV vaccination among adults eligible to be vaccinated and parents considering immunizing their children.

Strengths and Limitations

The study had several strengths. Since we used a nationally representative sample, the study findings are generalizable to all adult non-institutionalized US adults identifying as women. Another strength of this study was its longitudinal approach spanning over a decade and four datasets, allowing for the examination of changes in the relationship between HPV vaccination and cervical cancer prevalence across time. Additionally, utilizing NHANES, a publicly available dataset with standardized data collection methods, increased reproducibility and reliability.

However, the study encountered several limitations. A significant proportion of vaccinated participants fell within the age range of 20–39 years (88.73%). While cervical cancer typically develops after 50 years of age, the actual cancer development occurs 10–20 years after the initial HPV infection. Therefore, age is a major confounder, and since most participants were not yet at risk of developing cancers, the likelihood of detecting any association was potentially reduced. Excluding age and pregnancy status from the analysis resulted in a more pronounced association compared to the unadjusted model, with an odds ratio (OR) of 0.20 [95% CI: 0.03, 0.66] ($p = 0.03$). These findings suggest a temporal influence of age on the observed relationship. Furthermore, as HPV vaccination is recommended before the onset of sexual activity, typically around ages 11–12 years, older individuals may experience reduced effectiveness of the vaccine due to prior infection. Additionally, our study was unable to determine whether participants received vaccination before their first sexual encounter, as only a small fraction of respondents provided information on sexual behaviour. Therefore, we acknowledge the presence of temporality bias in our study.

Moreover, since one-dose vaccination might not provide full protection against HPV infections, it might then be inappropriate to compare at least one-dose vaccinated versus unvaccinated. Due to the limited number of cervical cancer cases ($n=98$), it was challenging to accurately estimate the effects of full and partial HPV vaccination compared to non-HPV vaccination. We also acknowledge that this rarity of observed effects resulted in insufficient statistical power to detect a statistically significant difference between the vaccinated and unvaccinated groups.

Therefore, we were unable to assess the potential effect modifications of covariates (e.g. age, race, pregnancy) on the relationship. Possible confounders like HIV status, number of children, lifestyle behaviours (e.g., smoking, healthcare access) were also not adjusted for, which could affect the relationship between HPV vaccination and cervical cancer.

Lastly, individuals opting for vaccination tend to engage in less risky behaviours and adopt more health-promoting habits, potentially skewing associations between vaccination status and the development of cancer, resulting in a healthy user bias. Those dying from cervical cancer could not be included, resulting in a survivor bias. Additionally, NHANES chooses participants based on household randomization and conducts interviews and surveys in the participants' households. Those currently not living in a residential household (e.g., homeless individuals living in shelters) were excluded from the study sample, resulting in a selection bias. Furthermore, vaccination status is self-reported and “Refused” ($n=3$) and “Don’t Know” ($n=431$) were considered to be missing values for vaccination status, which may have inadvertently excluded vaccinated individuals, introducing recall bias. However, recall bias may be mitigated to some extent as participants were specifically asked about receiving HPV vaccination under brand names Cervarix, Gardasil, or Gardasil 9.

Conclusion

The analyses suggest that HPV-vaccinated US women aged 20-59 years tend to have lower odds of developing cervical cancers than unvaccinated people. However, this difference is not statistically significant. Future research should link cancer registry and vaccination data for stronger evidence. It is understood that the effect of HPV vaccination will be observed in the next coming decades; however, preliminary findings, such as this study, could be used as additional evidence to encourage vaccination among 11- to 15-year-old adolescents and eligible adults to prevent HPV-related cancer cases in the future. These results may also be useful for policymakers to plan public health programs better to improve the uptake of HPV vaccines for vulnerable groups: Asians, low-income and high school graduates.

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Tables and Figures

Table 1: Sample population demographics of US women aged 20-59 years who did and did not get Human Papillomavirus vaccination between 2011 and 2020 from the National Health and Nutrition Examination Survey

Variable		Vaccinated	Unvaccinated	Sample Distribution (%)	p ³
n		1180	7416		
Age in years (Median [Q1, Q3]) ¹		25.0 [22.0, 31.0]	42.0 [33.0, 51.0]		<0.001
Race – n (%)	Asian	135 (11.4)	1078 (14.5)	1213 (14.11)	<0.001
	Black	310 (26.3)	1817 (24.5)	2127 (24.7)	
	Mexican American	125 (10.6)	1050 (14.2)	1175 (13.67)	
	White	410 (34.7)	2397 (32.3)	2807 (32.65)	
	Other, includes other Hispanic and multiracial	200 (16.9)	1074 (14.5)	1274 (14.82)	
Education – n (%) ²	12 th grade or less	115 (9.7)	1328 (17.9)	1443 (16.79)	<0.001
	High school graduate/GED or equivalent	198 (16.8)	1495 (20.2)	1693 (19.69)	
	Some college or AA degree	521 (44.2)	2515 (33.9)	3036 (35.32)	
	College graduate or above	346 (29.3)	2074 (28.0)	2420 (28.15)	
Income – n (%) ²	Under \$20,000	179 (15.2)	1105 (14.9)	1284 (14.94)	<0.001
	\$20,000 to \$54,999	241 (20.4)	1697 (22.9)	1938 (22.55)	
	\$55,000 to \$99,999	114 (9.7)	989 (13.3)	1103 (12.83)	
	\$100,000 and over	98 (8.3)	899 (12.1)	997 (11.50)	
Ever been pregnant – n (%) ²	Yes	561 (47.5)	5155 (69.5)	5716 (66.49)	<0.001
	No	451 (38.2)	1061 (14.3)	1512 (33.51)	

¹ Abbreviations: Q1, first quartile (25th percentile); Q3, third quartile (75th percentile). Percentages or median (first and third quartile) are shown as appropriate unless otherwise indicated. Median was reported due to the non-normal distribution of age.

² Missing: Education (0.04%), Income (38.09%), Ever been pregnant (15.91%)

³ Chi-square (χ^2) tests were employed for Race, Education, Income, and Ever been pregnant as both the outcome and dependent variables are categorical. For the variable Age, Wilcoxon Rank Sum test was applied due to its numeric nature and non-normal distribution, as assumptions for parametric tests were not met.

Table 2: Count and percentages of cervical cancer by vaccination status

Vaccinated for HPV	Cervical Cancer		
	Yes	No	Total
Yes	6 (0.5%)	1174 (99.5%)	1180
No	92 (1.2%)	7324 (98.8%)	7416
Total	98	8498	8596

Table 3: Logistic regression models investigating the association of Human Papillomavirus (HPV) vaccination history with cervical cancer cases among U.S. women aged 20-59 years in the National Health and Nutrition Examination Survey (NHANES), 2011-2020

Vaccination Status	Cervical Cancer			
	Model 1 OR (95% CI)	p-value	Model 2 OR (95% CI)	p-value
Unvaccinated (n = 7416)	Reference		Reference	
Vaccinated (n = 1180)	0.41 (0.16, 0.85)	0.033	0.4 (0.06, 1.47)	0.25

Abbreviations: CI: Confidence interval; n: sample size; OR: odds ratio

Model 1: unadjusted, bivariable regression model with variables vaccination status and cervical cancer.

Model 2: multivariable regression model, adjusting for all variables: age, education, race, income, and ever been pregnant.

Figure 1: Selection of final analytic sample using combined NHANES datasets and study design

*These values represent participants who completed the questionnaire (17) but may not have completed the physical examination. They were retained in the analytic sample as the outcome and explanatory variables were derived from the questionnaire. Consequently, this inclusion may lead to missing values in the covariates

